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Production and Optimization of Single-Cell Protein from Agricultural and Environmental Wastes: A Review

Allamin, I. A.[1*], Shettima, H.[2], Abba, R.Z.[3], Ismail, H.Y.[1], Abbator, A.[4] and Allamin, M.A.[5]

[1]Department of Microbiology,
Faculty of Life Sciences,
University of Maiduguri,
P.M.B. 1069, Maiduguri,
Borno State,
Nigeria.

[2]Department of Biological Sciences Faculty of Science,
Kashim Ibrahim University Maiduguri,
Nigeria
[3]Department of Food Science and Technology,
Faculty of Engineering University of Maiduguri,
P.M.B. 1069, Maiduguri,
Borno State,
Nigeria.

[4]Department of Botany,
Faculty of Life Sciences,
University of Maiduguri,
P.M.B. 1069, Maiduguri,
Borno State,
Nigeria.

[5]Center for Biotechnology,
University of Maiduguri,
P.M.B. 1069, Maiduguri,
Borno State,
Nigeria.

Email: ibnallamin@gmail.com

Abstract

Recent studies highlight the capacity of certain bacterial strains, particularly those in agricultural and environmental wastes, to utilize organic carbons as a sole carbon sources, converting them into high-value products like Single cell proteins (SCPs). Hence, the search of the bacteria in various environments, this review showed the potential of isolating SCP producing bacteria form both agro waste disposed soil and hydrocarbon polluted soils. The optimization of fermentation conditions, such as temperature, pH, and aeration in SCP production shows a significant progress made in providing the optimum condition for higher production of SCP from various bacterial isolates isolated from the mentioned environments. However, limited studies focus on agricultural waste substrates, where parameters must be tailored to microbial needs for optimal growth and protein synthesis. The

**Author for Correspondence*

biochemical composition, amino acid profile, and digestibility of SCP produced from agricultural waste remain poorly characterized. Studies have shown promising results for using microbial SCP as protein sources in animal feed, yet the effects of agricultural waste composition on these characteristics need further investigation. While laboratory-scale studies have demonstrated the feasibility of SCP production from agricultural and environmental wastes, scaling up to industrial-scale presents significant challenges related to process control, cost-effectiveness, and regulatory compliance.

Keywords: Single Cell Proteins; Agricultural; Environment; Waste and Production.

INTRODUCTION

Single Cell Protein (SCP) refers to the dried biomass of microorganisms, including algae, fungi, and bacteria, that can be utilized as a protein source for human and animal consumption. The term was first coined in the 1960s by Carol L. Wilson at the Massachusetts Institute of Technology (MIT), highlighting the potential of microbial cells to serve as a rich protein source, often comprising 60-82% of dry cell weight in protein content (Abdullahi *et al*., 2021). SCP is recognized not only for its high protein levels but also for its additional nutritional components, including fats, carbohydrates, vitamins, and minerals, making it a versatile dietary supplement (Obaida, 2021). The utilization of SCP is particularly relevant in addressing global protein shortages and enhancing food security, especially in regions with limited agricultural capacity (Kumar, 2023).

The exploration of SCP began in earnest during the mid-20th century, driven by the increasing demand for protein sources amidst rising global populations and food scarcity concerns. The first significant studies on SCP emerged in the 1960s, focusing on the cultivation of yeast and bacteria for protein production (Abdullahi *et al*., 2021). Over the decades, research has expanded to include various microbial species, cultivation techniques, and applications in food and feed industries. This historical evolution reflects a growing recognition of SCP as a sustainable alternative to traditional protein sources, particularly in light of environmental concerns associated with livestock farming (Abodunde, 2023).

SCP plays a crucial role in the contemporary food system by providing an alternative protein source that can be produced with lower environmental impact compared to conventional animal protein sources. The production of SCP can utilize agricultural and food waste, thus contributing to waste management and sustainability (Kumar, 2023). Moreover, SCP is characterized by rapid growth rates and high feed conversion efficiencies, making it an attractive option for meeting the protein needs of a growing global population (Uysal *et al*., 2002). The nutritional profile of SCP, which often includes essential amino acids lacking in many plant-based proteins, further enhances its importance as a dietary supplement (Obaida, 2021).

Single cell protein from microorganism

Single Cell Proteins (SCP) are derived from various microorganisms, including bacteria, yeasts, fungi, and algae (Table 1) and are recognized for their rich nutritional profile. SCP serves as a sustainable protein source, addressing global food security challenges (Kumar, 2023). The nutritional constituents of SCP include essential amino acids, fatty acids, vitamins, minerals, and nucleic acids, making it a valuable dietary supplement. Yeasts Particularly species like *Saccharomyces cerevisiae*, which are widely used in food production and fermentation processes. Yeasts are known for their high protein content and ability to grow on various substrates, including waste materials (Obaida, 2021). Yeast which are unicellular fungi constitute among the diverse form of microbes with the potential to produce SCP when

subjected to most readily. Certain bacteria, such as *Methylophilus methylotrophus*, can be cultivated on methanol or other substrates, yielding high protein content suitable for animal feed (Abdullahi *et al.,* 2021). Mycoprotein, derived from fungi like *Fusarium venenatum*, is a popular SCP used in meat substitutes due to its fibrous texture and nutritional profile (Spiegel *et al*., 2020). Species such as *Spirulina* and *Chlorella* are rich in protein and other nutrients, making them valuable for both human consumption and aquaculture (Abodunde, 2023). Each type of SCP has distinct advantages, including rapid growth rates, high protein yields, and the ability to utilize non-arable land for cultivation, thus contributing to food security and sustainability (Kumar, 2023).

Table 1: Percentage SCP produce by various microorganisms

Microorganisms	SCP (%)
Yeast	30-60
Bacteria	40-80
Fungi	30-65
Algae	30-75

Source: (Zhuang *et al.,* 2024)

Nutritional Composition of SCP

Amino acid Approximately 80% of SCP's nitrogen content is composed of amino acids, comparable to traditional protein sources (Ogbonna *et al.,* 2013). Fatty Acids in SCP contains beneficial fatty acids, contributing to its overall nutritional value (Abodunde, 2023). Vitamins and Minerals SCP is rich in vitamins (e.g., B vitamins) and trace minerals, enhancing its role as a dietary supplement (Matassa *et al.,* 2016; Abodunde, 2023). Nucleic Acids in SCP has a higher nucleic acid content than conventional proteins, which can pose a risk of allergic reactions in some individuals (Abodunde, 2023). The percentage composition is shown in table 2.

Table 2: Percentage nutrients in SCP

Nutrient	Composition (%)
Crude Protein	30-80
Carbohydrates	10-30
Fatty Acids	15-20
Vitamins and Minerals	10-20
Nucleic acid	2-15

Source: (Obaida, 2021).

Advantages of Single Cell Protein

The advantages of SCP over traditional protein sources are manifold:

i. Sustainability: SCP production can significantly reduce the environmental footprint associated with animal farming, including greenhouse gas emissions, land use, and water consumption (Abdullahi *et al.,* 2021).
ii. Nutritional Value: SCP is often rich in essential amino acids, vitamins, and minerals, making it a nutritionally dense option compared to many plant-based proteins (Obaida, 2021).
iii. Rapid Production: Microorganisms can be cultivated in controlled environments, allowing for rapid biomass production compared to traditional crops or livestock (Uysal et al., 2002).
iv. Utilization of Waste: SCP can be produced from agricultural and food waste, contributing to waste reduction and circular economy practices (Satpute et al., 2010; Kumar, 2023).
v. Versatility: SCP can be incorporated into various food products, animal feeds, and dietary supplements, providing flexibility in its application (Abdullahi et al., 2021).

Disadvantages of Single Cell Protein

Despite its numerous advantages, SCP also presents certain challenges:

i. Acceptance and Sensory Attributes: Consumer acceptance of SCP, particularly from fungal or bacterial sources, can be hindered by sensory attributes and perceptions of microbial food sources (Grasso et al., 2019).
ii. Potential Allergens: Some SCP products may pose allergenic risks, necessitating thorough safety evaluations before widespread adoption (Spiegel et al., 2020).
iii. Production Costs: While SCP can be produced sustainably, the initial setup and operational costs for large-scale production facilities can be significant, potentially limiting accessibility (Grasso *et al.,* 2019).
iv. Nutritional Imbalance: Depending on the source, SCP may lack certain nutrients or amino acids, necessitating careful formulation when used as a dietary supplement (Obaida, 2021).
v. Regulatory Hurdles: The approval process for new food sources, including SCP, can be complex and time-consuming, impacting market entry and consumer availability (Spiegel *et al.,* 2020).

Production and Sustainability of SCP

Renewable resources are mostly used nowadays to produces SCP importantly like agricultural and industrial waste, promoting sustainability (Matassa *et al.,* 2016; Abodunde, 2023). It also has nonstop accessibility (Year-Round Production), Unlike traditional crops, SCP can be produced continuously, providing a reliable protein source (Mahmud et al., 2022). The synthesis of SCP from bacterial species has drawn interest recently as a novel and sustainable solution to the world's need for high protein. From strain selection to distributing the SCP products to final consumers, the SCP production process typically consists of multiple steps (**Figure 1**). After harvesting, various cell disruption techniques are used in the protein recovery process to liberate intracellular proteins. With an emphasis on effectively releasing intracellular proteins, techniques including mechanical (ultrasonication, bead milling, high-pressure homogenization, and microfluidization), chemical (alkaline treatment, acid treatment, and chaotropic agents), and biological (enzymes like lysozyme, cellulase, and proteases). Following recovery, the proteins are concentrated for animal feed by dehydration, precipitation, ultrafiltration, spray drying, and lyophilization (Abodunde, 2023). However, following further procedures to eliminate nucleic acid from SCP products becomes crucial when using them as a protein substitute for human consumption. The nucleic acids are removed using traditional techniques such heat shock, acid treatment, and base treatment. The nucleic acids in the SCP product are also being eliminated by hydrogen peroxide treatment and enzymatic action. Every stage needs to be tuned for high yields and cost-effective SCP appropriate for applications (Hadi & Brightwell 2021). As was previously said (Spalviņš *et al.,* 2020), numerous bacterial species have demonstrated significant potential among the many types of bacteria used to make SCP because of their distinct metabolism of C1 carbon rich chemicals as a primary energy and carbon source. According to reports, bacterial strains generate phospholipids, vitamins, minerals, and a very high crude protein content (70–80%) that may be used in various feedstocks. Depending on the strains, the protein content in certain species can vary between 50 and 80 percent. *Pseudomonas, Bacillus, Acinotobacter, and Methylophilus* are the most often used bacterial taxa in the manufacture of SCP (Abdullahi *et al.,* 2021). Researchers from all around the world have attempted to use bacterial species to their fullest capabilities in order to produce SCP (Simões *et al.,* 2022). The primary goal of research has been to optimize the growing conditions in order to maximize the protein content or specific amino acids. SCP has been made using a variety of fermentation techniques, including fed-batch, continuous, and batch procedures. Based on the selection of

microorganisms for SCP production and process efficiency, each mode of operation has been optimized (Zhuang *et al.,* 2024). In order to produce SCP with methane as the substrate, optimized the fed-batch process using response surface methodology (RSM) and Plackett-Burman Design (Zhuang *et al.,* 2024).

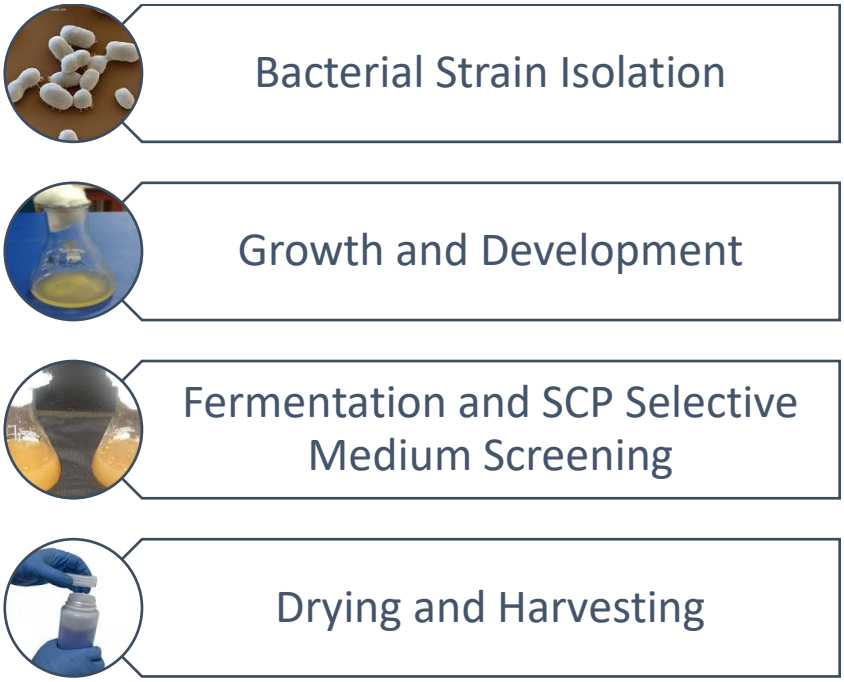


Figure 1: Single cell protein (SCP) production process

SCP from Agro-industrial Wastes

Agro industrial waste are waste from the environment and agricultural products. Such waste constitute rich source of carbon required for SCP production by microbes. Potato starch production produces a substantial amount of trash, which poses a serious environmental problem. Potato pulp's low nutritional value and high water content typically more than 80% limit its use in animal feed. Simões *et al.,* (2022) examined a novel method for addressing solid and liquid pollution in the starch sector that used the combined fermentation of potato residue and wastewater without the use of additional nitrogen sources. Four important microorganisms *Curacaobacter, Pseudoalteromonas, Paenibacillus,* and *Bacillus* were found to be involved in this process. Protein made up 46.09% of the final SCP products. Scaling up to a 150 m^3 fermentation system showed the potential for industrial applications. Furthermore, the production of high-quality SCP has shown significant promise when waste materials from date syrup and cheese leftovers are used as substrates. The main components of lignocellulosic biomass, which is produced by photosynthesis in plants, are cellulose, hemicellulose, and lignin. Its entire potential is still mostly unrealized despite being used in a number of industries, such as papermaking, building, textiles, and wood processing. For the two main methods of producing SCP enzymatic hydrolysis followed by yeast fermentation or simultaneous saccharification and fermentation employing cellulose and hemicellulose-degrading bacteria and yeast the bioconversion of lignocellulosic biomass is crucial.

The most common bioresources for bioproducts, lignocellulosic hydrolysates, have been shown to be efficiently utilized by yeast species. (Zhuang *et al.,* 2024) optimized the nutritional conditions for cultivating *Paecilomyces variotii* in eucalyptus hemicellulose hydrolysate by adding ammonium sulfate and sodium phosphate, resulting in a cell concentration of 12.06 g/L. They used lignocellulosic hydrolysates and xylose to generate SCP with *Candida*

intermedia FL023, using peptone and yeast extract as nitrogen sources. They achieved a crude protein content of 484.2 g/kg DCW. These studies highlight the potential of lignocellulosic biomass and a variety of agro-processing wastes as sustainable substrates for SCP manufacture, promoting both economic viability and environmental sustainability across several industrial sectors (Zhuang *et al.,* 2024). The global production of paper and cardboard reached 390 million tons in 2018 (Kriss *et al.,* 2018). At present, in developed and developing countries including Europe, North America and Africa the average rate of paper recycling equals 60–70 % of the total output. However, in certain developed and developing countries the rate of paper recycling is much lower and the production of paper is expected to reach 500 million tons a year in 2030 (Kriss *et al.,* 2018). Therefore, the available amount of waste paper is very large and it is expected that in the future it will continue increase. The main component of paper waste is cellulose. Hydrolyzed paper waste contains about 60–70 % sugar, of which about 69 % is glucose, 21 % xylose, 3.5 % mannose, 2.5 % arabinose, 1.5 % galactose, and other 35–45 % consists of 20 % lignin, 60 % clay mineral kaolinite, and other residues (Ivarson & Morita 1982). Generally, SCP production from waste paper is laden by contamination of unwanted microbial material. Thus maintenance of contamination increases the total cost of production. In order to circumvent these costs, a good substitute is the use of extremophilic microorganisms in the production of SCP, therefore selective growth conditions could be kept without the risk of contamination.

SCP from Environmental Wastes

Hydrocarbons, which are generated from petroleum refineries, can serve as an alternative to more conventional carbohydrates. Unlike carbohydrates, hydrocarbons have no oxygen and are practically insoluble in water. The use of petroleum residues in SCP production provides an opportunity to reduce the environmental impact caused by the oil industry, although, in the fermentation process only a small proportion of hydrocarbons is used as feedstock. SCP producing microorganisms can use fuel oil and other n-parafins, which are distillates or residues from petroleum distillation. Studies highlight the capacity of certain bacterial strains, particularly those from hydrocarbon-contaminated environments, to utilize hydrocarbons as a sole carbon source, converting them into high-value products like SCPs (Youssef *et al.,* 2004; Kriss *et al.,* 2018). The optimization of fermentation conditions, such as temperature, pH, and aeration, has been widely studied in SCP production from plant-based wastes. However, limited studies focus on hydrocarbon substrates, where parameters must be tailored to microbial needs for optimal growth and protein synthesis (Yang, 2024). The biochemical composition, amino acid profile, and digestibility of SCPs produced from hydrocarbons remain poorly characterized. Studies show promising results for using microbial SCPs as protein sources in animal feed, yet the effects of hydrocarbon waste composition on these characteristics need further investigation. While laboratory-scale studies (Matassa *et al.,* 2016; Ime *et al.,* 2020) have demonstrated the feasibility of SCP production from hydrocarbon waste, scaling up to industrial levels presents significant challenges related to process control, cost-effectiveness, and regulatory compliance. In areas where latex-rich plants are grown, rubber sheet production is highly prevalent. Wastewater from the production of latex rubber is high in ammonia, formic acid, sodium sulfite, and sodium metabisulfite. These wastewaters must be treated before being released into nearby water bodies because they pose a risk to the environment. Wastewater from latex rubber is currently treated using lagoons and oxidation ponds, but these methods are inefficient because they do not fully oxidize the compounds in the wastewater. When these simple treatment systems are used, greenhouse gases like carbon dioxide and methane are released into the atmosphere along with an unpleasant smell. Therefore, specialized fermentation of this wastewater may be a good substitute for the existing technologies. According to reports, using wastewater from latex rubber sheets to

produce SCP can result in significant and high protein concentrations in the microbial biomass.

Methane has long been researched as an inexpensive and accessible carbon source for the synthesis of SCP. Methane has advantages in the production of SCP due to its selectivity, low toxicity, and volatility, despite being extremely flammable, poorly soluble, and naturally occurring with various impurities. Natural gas is thought to be one of the best sources of methane for the production of SCP because of its high methane content (85–90%). Thus far, using natural gas to cultivate experiments has produced biomass of microorganisms with extremely high SCP concentrations. The natural gas industry wastes more than 16 billion cubic meters of natural gas every year in the United States alone as a result of venting, flaring, and leaks in its operations. In addition to causing enormous financial losses, this massive amount of wasted gas releases enormous amounts of methane.

SCP production optimization

Optimizing the production of single-cell proteins (SCP) from bacterial sources involves a multifaceted approach that considers various agro-industrial environmental and nutritional parameters. Key factors influencing SCP production include temperature, pH, aeration, and carbon source concentration.

Temperature optimization

Temperature is a critical factor affecting microbial growth and protein synthesis. Each bacterial strain has an optimal temperature range that maximizes growth rates and metabolic activity. For instance, *Bacillus subtilis* has been shown to exhibit significant variations in protein production at different temperatures, with optimal ranges typically between 30°C and 37°C (Parthipan *et al.,* 2017). Similarly, studies on *Streptomyces* species indicate that temperature adjustments can lead to enhanced biosynthesis of secondary metabolites, which are often linked to protein production (Yang, 2024). Therefore, determining the specific thermal preferences of the bacterial strain in question is essential for maximizing SCP yields.

pH optimization

The pH of the growth medium significantly influences enzyme activity and microbial metabolism. Most bacteria thrive in neutral to slightly alkaline conditions (pH 6.5 to 8.5). For example, research indicates that an alkaline shift in pH can enhance biomass and protein production in *Escherichia coli* when acetate is used as a carbon source; composition of substrate used from both agro-industrial and environmental waste affect the pH hence has direct effect on the bacterial growth (Leone *et al*., 2015). Furthermore, the optimization of pH has been linked to improved enzyme production, which is crucial for the breakdown of substrates form agricultural and industrial wastes into usable forms for microbial growth (Kłosowski *et al*., 2018). Therefore, conducting pH optimization experiments is vital to identify the optimal range for SCP production.

Aeration and Agitation

Aeration and agitation are crucial for maintaining dissolved oxygen levels, which are essential for aerobic bacteria. Studies have shown that increasing aeration rates can enhance the growth of bacteria such as *Methylobacterium organophilum*, thereby improving SCP and metabolite production (Simões *et al*., 2022). The agitation speed also plays a role in nutrient distribution and oxygen transfer within the culture medium, impacting overall microbial growth and protein synthesis; hence the composition of the substrate is not directly affecting the microbial requirement to aeration (Raita *et al*., 2022). Therefore, optimizing both aeration and agitation

conditions is necessary to ensure efficient microbial respiration and nutrient uptake as environmental condition not directly on the substrate used.

Carbon Source Concentration

The choice and concentration of carbon sources are pivotal in determining the metabolic pathways utilized by bacteria, which directly affects SCP yield. High concentrations of carbon sources can lead to substrate inhibition, while insufficient concentrations may result in starvation and reduced growth rates, Substrate form both agro-industrial and environmental sources are screen base on the rich composition of carbon in them (Kumar *et al*., 2019). For example, studies indicate that a balanced carbon-to-nitrogen ratio is crucial for optimal growth and protein production in various bacterial strains (Spalviņš *et al.,* 2020). Additionally, the use of alternative carbon sources, such as plant stalks, fruit peels, and animal waste, has been explored to enhance SCP production while reducing costs (Al-Obeidi, 2023). Thus, careful selection and optimization of carbon source concentrations are essential for maximizing SCP yields.

Safety and Nutritional Value Assessment

Single-cell proteins (SCPs) have emerged as a promising alternative protein source for animal feed and human consumption, primarily due to their high nutritional quality and sustainability. SCPs are derived from microorganisms such as bacteria, yeast, and fungi, which can be cultivated on various substrates, including agricultural waste and food by-products. This cultivation process not only provides a rich source of protein but also addresses environmental concerns related to traditional protein production methods.

One of the key advantages of SCPs is their high protein content, which can range from 60% to 80% dry matter, making them a potent source of dietary protein for both animals and humans (Williams *et al.,* 2023; Kumar, 2023). For instance, studies have shown that SCPs derived from yeast and bacteria can effectively replace fish protein hydrolysate in aquaculture diets, leading to positive growth outcomes in species such as zebrafish (Williams *et al.,* 2023). This indicates that SCPs can serve as a viable alternative to conventional protein sources, potentially reducing reliance on overfished marine resources.

In terms of nutritional quality, SCPs are not only rich in protein but also contain essential amino acids, vitamins, and minerals. The bioactive components present in SCPs can enhance gut health and improve nutrient absorption in animals (Elghandour *et al.,* 2019; Cheng & Kim, 2022). For example, the inclusion of SCPs in the diets of nursery pigs has been shown to enhance energy utilization and promote beneficial gut microbiota, which is crucial for optimal growth and health (Cheng & Kim, 2022). Additionally, SCPs can contribute to the production of short-chain fatty acids through fermentation processes in the gut, further supporting animal health (Cheng & Kim, 2022).

However, the safety of SCPs is a critical consideration. The production of SCPs must be carefully monitored to mitigate risks associated with microbial contamination, toxins, and allergens (Hadi & Brightwell, 2021). Therefore, rigorous safety assessments and regulatory frameworks are essential to ensure that SCPs are safe for consumption and do not pose health risks to animals or humans.

CONCLUSION

As an alternate source of protein, SCP is particularly relevant in addressing global protein shortages and enhancing food security, especially in regions with limited balanced dietary access. By utilizing environmental and agricultural waste and by-products, SCP production can significantly reduce the environmental footprint associated waste management. This aligns with global efforts to promote sustainable food systems and reduce greenhouse gas emissions from animal agriculture. Studies indicated the potential some microorganisms used on large quantity of agro-industrial and environmental waste to produce SCP.

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